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# Evaluating Model Performance for Inter-County Net Migration Patterns in the United States

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## 1. Introduction

Inter-state and inter-county migration patterns within the United States are of keen interest to many different stakeholders, including property owners, local, state, and federal governments, and all residents. Shrinking counties face many challenges, including reduction in economic activity, falling property values, and underfunding. Meanwhile, growing counties face their own set of challenges, such as increasing rents, overwhelmed infrastructure, and traffic. It is therefore highly valuable to understand what factors lead people to move from place to place, and to predict where migration hot spots will be. This information can be used to optimize decision making regarding tax rates, policy and infrastructure, as well as personal planning.

The University of Wisconsin compiles this data in the '[Net Migration Patterns for US Counties](#)' tool. This dataset includes population and net migration numbers, broken down both by age (in 5-year buckets) and race, for all 3,143 counties in the United States. Additionally, the tool includes the ability to append additional data for each county (Table 1). Demographic data from the 2010 Census includes racial and age category breakdowns and housing data. Each county also has certain associated designations by the United States Department of Agriculture known as Economic Research Service (ERS) county economic typology codes. These categorizations provide insight into the economic conditions of each county, such as dominant industries, poverty rates, and education levels (Parker 2015).

Table 1. Summary of Features

| Feature                  | Year | Feature                     | Year    |
|--------------------------|------|-----------------------------|---------|
| % Hispanic               | 2010 | Metro/micro/non-core        | 2020    |
| % households w/ children | 2010 | Farming-dependent           | 2015    |
| Housing units            | 2010 | Mining-dependent            | 2015    |
| % male                   | 2010 | Manufacturing-dependent     | 2015    |
| Median age               | 2010 | Government-dependent        | 2015    |
| % Asian                  | 2010 | Recreation-dependent        | 2015    |
| % Black                  | 2010 | Nonspecialized              | 2015    |
| % White                  | 2010 | Low education               | 2015    |
| % 65 plus                | 2010 | Low employment              | 2008–12 |
| % owner occupied         | 2010 | Population loss             | 2010    |
| Population               | 2010 | Retirement destination      | 2015    |
| % seasonal               | 2010 | Persistent poverty          | 2013    |
| % single parent          | 2010 | Persistent child poverty    | 2013    |
| % under 18               | 2010 | Rural-urban continuum codes | 2023    |
| % vacant                 | 2010 |                             |         |

Recognizing demographic shifts and predicting trends has been the subject of numerous studies and analyses. For example, a study from Mississippi State University analyzed the upward trend in the median age of Mississippi residents and the potential consequences to tax revenues, budget allocation, and the labor market (Kim & Mills 2024). Inter-county migration is a key driver of these demographic changes, and as a result, has been studied extensively. A 2015 paper by Johnson et al. found particular "migration signatures" by plotting migration by age and race against counties categorized as metro, suburban, small, or rural (Johnson & Winkler 2015). The authors identified age-specific patterns that held consistent through the decades, with large metro areas seeing an influx of people in their twenties, and suburbs seeing large influxes of people in their thirties. A study by the Brookings Institute utilized this data to analyze the migration of Black Americans, finding that Southern states saw significant out-migration of Black Residents in the 1960s, with the trend reversing in more recent decades (Frey 2004).

There have been many attempts to model migration mathematically. The Gravity Model, introduced in 1941, repurposes the formula for gravitational potential energy and treats two locations as physical objects, estimating migration between them using their populations and the distance between them (Stewart 1941). A 2016 case study on internal migration within New Zealand argued that the gravity model is still a useful tool for modeling migration even in the modern age, but raised concerns about its applicability in an increasingly interconnected world and with the increased availability of data (Poot et al. 2016). A 2025 study used this statistical model among others to assess Internal Revenue Service (IRS) data, the American Community Survey, and Census Long Form Data for inter-county migration (Morefield & Leslie 2025). This study elucidated the trade-offs between using these different data sets for different modeling purposes. The impact of economic and political factors was also explored in a 2024 study, which utilized ERS Typology data to build a logistic regression model. The authors found that rural counties had the highest propensity for out-migration (Xu 2024).

We present a comprehensive review of different machine learning models (including both regression and classifica-

tion), building off of the prior demographic and migration research. We utilized the migration and census data, as well as the ERS typology classifications curated by the University of Wisconsin and optimize for highest performance. We then provide an assessment of model utility for predicting net migration for each county in the United States. By selecting the highest performing models and analyzing the feature weights, we can then determine which demographic, geographic, and economic factors play the largest role in motivating people to migrate to or away from a given county.

## 2. Data

Each observation in the dataset represents a U.S. county. In the dataset, each county is treated as an independent observation. The models predict overall net migration rate per county rather than migration exchanges between specific pairs of counties. After preprocessing, our dataset contains 3,126 U.S. counties and 29 features containing demographic and economic characteristics that contribute to changing migration rates (Table 2). Our target variable is net migration, a continuous variable representing the difference between in-migration and out-migration in a county per 1,000 residents. The mean for the target variable is approximately 0, meaning that on average, in-migration and out-migration are balanced across counties, with a standard deviation of 8.56. The range for this target variable is -35.40 to 92.62 (Table 2). The distribution shows that values are centered around zero with a right-skewed distribution and some extreme positive values (Figure 2). While most counties experience small migration changes, a small number of counties experience much higher positive migration rates.

Feature interactions between scaled variables help us understand relationships between variables and identify potential multicollinearity concerns in the dataset (Figure 1). Several demographic and economic variables exhibit moderate correlations with each other.

Somerset County, NJ serves as an example row in our training data (Table 3). Each county has relevant information containing meaningful demographic data, total population statistics, and binary economic indicators pertaining to whether a county experiences dependence on government, farming, manufacturing, or other typologies for economic activity.

For classification, the target variable was converted to a binary variable, where counties with a positive net migration rate received a label of 1 and counties with a negative net migration rate received a label of 0. The distribution is relatively balanced, with approximately 54 percent of counties experiencing negative migration rates and 46 percent of counties experiencing positive migration rates.

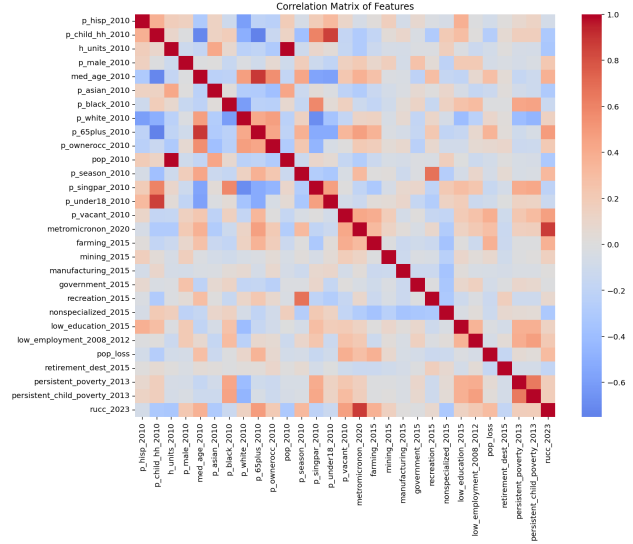


Figure 1. Correlation Matrix of Features

Table 2. Summary statistics for the target variable.

| Statistic          | Value  |
|--------------------|--------|
| Count              | 3126   |
| Mean               | 0.003  |
| Standard Deviation | 8.562  |
| Minimum            | -35.40 |
| 25th Percentile    | -4.77  |
| Median             | -0.67  |
| 75th Percentile    | 4.18   |
| Maximum            | 92.62  |

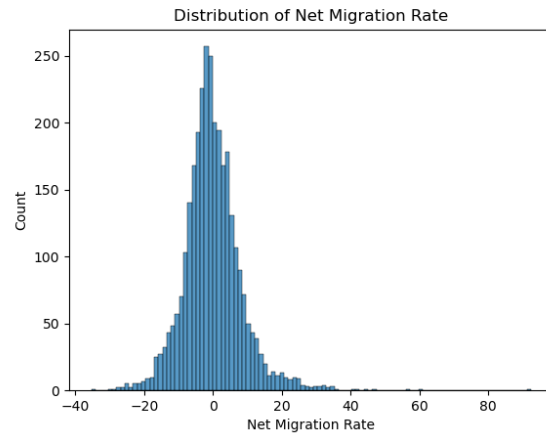


Figure 2. Distribution of Net Migration Rate

## 3. Methods

### 3.1. Data Cleaning

For data preprocessing, the datasets were loaded and merged using the county identifiers 'fips'. Column names were mapped to readable labels using the header file for better interpretability of results. A new dataframe was created

Table 3. Example Row of Dataset (Somerset County, NJ)

| Feature               | Value   | Feature            | Value   | Feature                   | Value       |
|-----------------------|---------|--------------------|---------|---------------------------|-------------|
| % Hispanic (2010)     | 13.00   | % Owner Occupied   | 76.80   | Recreation Dep.           | 0.00        |
| % Households w/ Child | 37.90   | Total Pop (2010)   | 323,444 | Nonspecialized Econ       | 1.00        |
| Housing Units         | 123,127 | % Seasonal Housing | 0.60    | Low Education             | 0.00        |
| % Male                | 48.70   | % Single Parent    | 6.10    | Low Employment            | 0.00        |
| Median Age            | 40.20   | % Under 18         | 25.00   | Population Loss           | 0.00        |
| % Asian               | 14.10   | % Vacant Housing   | 3.80    | Retirement Dest.          | 0.00        |
| % Black               | 8.50    | Metro/Micro/Non    | 1.00    | Persistent Poverty        | 0.00        |
| % White               | 62.40   | Farming Dependent  | 0.00    | Pers. Child Poverty       | 0.00        |
| % 65 Plus             | 12.40   | Mining Dependent   | 0.00    | RUCC (2023)               | 1.00        |
| Manufacturing Dep.    | 0.00    | Government Dep.    | 0.00    | <b>Net Migr. (Target)</b> | <b>3.83</b> |

by selecting chosen features and the target variable and dropping missing values. Next, the data was split into an 80-20 training test split and was standardized using `StandardScaler` to ensure that variables with larger numeric ranges did not have a disproportionately large effect on the model training.

Two different prediction tasks were considered for this project. Regression was used to predict the continuous migration rate for each county and classification was used to predict whether a county experienced a positive or negative net migration. Using both regression and classification tasks allow us to evaluate migration patterns at both continuous and binary levels.

### 3.2. Regression

In our regression models, we included models to help capture both linear and nonlinear relations. First, a baseline regression model was created using `DummyRegressor`, which predicts the mean migration rate for all counties for the purpose of having a baseline comparison for all other models.

#### 3.2.1. LINEAR

Our linear models assume a linear relationship between the features and target variable. These linear models include linear regression, stochastic gradient descent (SGD), ridge, lasso, and elastic net regression. The linear regression model is an unregularized ordinary least squares regression model. It aims to find the best fitting line through the data points and minimize the sum of squared residuals between each data point and the line. The SGD regression model takes the gradient of the loss function and updates the initial weights using one sample at a time. It adjusts the weights slightly based on the error and repeats this process for several data points to minimize the mean squared error (MSE). The ridge regression model uses an L2 penalty ( $\alpha = 1.0$ ) on large coefficients to reduce overfitting. Similarly, the lasso regression uses an L1 penalty ( $\alpha = 1.0$ ) on large coefficients that shrinks some coefficients to zero to reduce

overfitting. Elastic net regression combines lasso and ridge regression by incorporating the L1 and L2 penalties into the loss function. Elastic net regression is useful when there are a large number of features and helps handle potential issues of multicollinearity.

#### 3.2.2. NONLINEAR

To capture more complex relationships, we also implemented nonlinear models including decision trees, random forest, and gradient boosting. Migration patterns are influenced by a complex interaction between features, as seen in Figure 1, motivating the use of nonlinear models. The nonlinear models allow for interactions between variables, and are highly flexible and able to capture nonlinear relationships to fit the data more accurately. Our decision tree regression model splits the data into smaller subsets to search for the optimal split point within a tree. This process is repeated until the default stopping condition is met, where splits continue until nodes contain too few samples or no further reductions can be made to the MSE. The random forest regression model combines multiple decision trees during training and reduces overfitting by averaging multiple trees. Lastly, gradient boosting regression works by sequentially building models, typically using decision trees, to improve upon the errors of the previous trees. The errors are minimized through calculating residuals and training new models to fit them.

### 3.3. Classification

A baseline classification model was created using `DummyClassifier`, which predicts the majority class and serves as a benchmark for evaluating classification performance. A logistic regression model was implemented as a linear classifier to predict the probability of positive migration. A support vector machine (SVM) was used to identify an optimal decision boundary between the classes that maximizes the margin. Classification models of decision tree, random forest, and gradient boosting were performed as well to compare both linear and nonlinear classification

approaches.

### 3.4. Plots and Tables

We plotted the regression and classification metrics, including  $R^2$ , MSE, F1 score, and ROC-AUC to compare model performance across different approaches. We measured the regression models with  $R^2$  because it effectively provides a metric for measuring the model’s ability to handle variance of data, and MSE because it represents how poorly the model predicts. For the classification models, we measured F1 score because it provides a balanced read of the model’s accuracy and precision and ROC-AUC because it provides a measure of the model’s ability to classify between positive and negative correctly. Additionally, actual vs predicted and residual plots were used to assess model fit and identify prediction errors. Feature importance was also analyzed to understand the influence of different variables. For the classification models, ROC curves were plotted to evaluate the performance across different thresholds. These visualizations allow a qualitative assessment of model behavior.

### 3.5. Hyperparameter Tuning

We selected the two highest performing regression models (gradient boosting regression and random forest regression) and single highest classification model (gradient boosting classification) for hyperparameter tuning. Hyperparameter tuning aims to find the parameters to the model that will yield its best possible performance on the data. In order to compare hyperparameter tuning approaches as well as models, we implemented a Grid Search and Random Search for each of these three models. We recalculated  $R^2$  values for the regression models and ROC-AUC values for the classification model in order to observe performance changes following hyperparameter tuning. We chose to use `n_estimators`: [100, 200] in all searches in order to keep comparison fair and avoid providing one search modality more estimators, and chose 100 and 200 in order to strike a balance between reasonable search times and accuracy (Table 4). Since random search is generally faster, we gave it a wider range of options for estimators. For grid search in the gradient boosting models, we chose learning rates at 0.01 and 0.1 in order to provide one option at a standard level of caution (0.01), as well as another option at a faster but potentially more risky learning rate. For random search, we provided a larger set of learning rate options since runtime is less of concern. We provided the gradient boosting grid search with `max_depth` of 3 and 5, and gave random forest `max_depth` options of **None**, 10, and 20 because it is commonly observed for random forest optimal `max_depth` to be higher than gradient boosting optimal `max_depth`. We left **None** as an option for random search due to its expected faster run times. We gave the random forest regression `max_features` options of `sqrt` and `log2` accord-

ing to common practice and added 0.3 and 0.5 as options for random search to try certain percentages of features since there is no runtime downside. `min_samples_split` and `min_samples_leaf` are standard parameters used in random forest hyperparameter tuning. We picked one of them (`min_samples_split`) and provided it to the random forest grid search, but kept it to two options and did not provide it at all to gradient boosting grid search out of caution for runtime. We provided three options for `min_samples_split` for both random searches and provided three `min_samples_leaf` options for random forest random search since it is generally not used for gradient boosting. Similarly, `subsample` is a gradient boosting standard parameter, so we gave grid search a couple of commonly used options for it, and gave random search a similar wider range.

Table 4. Hyperparameter search configurations.

| Model      | Param.                | Grid                                  | Random                                           |
|------------|-----------------------|---------------------------------------|--------------------------------------------------|
| GB Reg/Clf | <code>n_est.</code>   | 100, 200                              | 50–300                                           |
|            | <code>lr</code>       | 0.01, 0.1                             | 0.005–0.2                                        |
|            | <code>max_d</code>    | 3, 5                                  | 2–6                                              |
|            | <code>subsamp.</code> | 0.8, 1.0                              | 0.6–1.0                                          |
|            | <code>min_spl.</code> | —                                     | 2, 5, 10                                         |
| RF Reg.    | <code>n_est.</code>   | 100, 200                              | 50–500                                           |
|            | <code>max_d</code>    | None, 10, 20                          | None, 5–30                                       |
|            | <code>max_f.</code>   | <code>sqrt</code> , <code>log2</code> | <code>sqrt</code> , <code>log2</code> , 0.3, 0.5 |
|            | <code>min_spl.</code> | 2, 5                                  | 2, 5, 10                                         |
|            | <code>min_lf.</code>  | —                                     | 1, 2, 4                                          |

## 4. Results

We took two different approaches to modeling inter-county migration. The first was to generate regression models to predict the values of net migration to each county based on our features. For this section, we ran Linear Regression, Ridge Regression, Lasso Regression, SGD Regression, Random Forest, Gradient Boosting, Decision Tree, and Elastic Net. We quantified the utility of these models with  $R^2$  and MSE. The second approach was to generate classification models to predict whether a county sees positive or negative net migration. For this section, we ran Decision Tree, Random Forest, Logistic Regression, SVM, and Gradient Boosting. We quantified the utility of these models with ROC-AUC and F1. We also plotted ROC curves for all classification models.

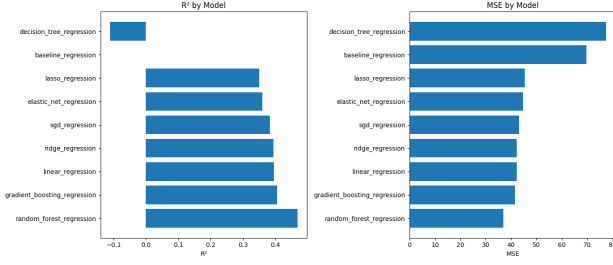


Figure 3. Regression model performance

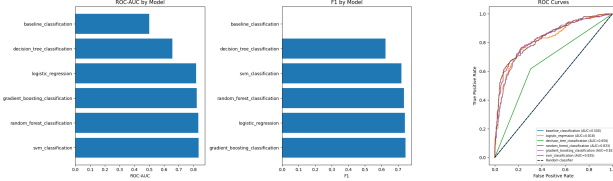


Figure 4. Classification model performance

The regression results show all models other than decision tree regression outperform the baseline regression (Figure 3). Gradient boosting and random forest regression were the two highest performing models, as evidenced by the highest  $R^2$  and lowest MSE values. We selected these two models for hyperparameter tuning. Linear regression (regularized and unregularized), SGD, and elastic net models all substantially outperformed the baseline.

#### 4.1. Classification Results

The classification results show all models outperform the baseline classification (Figure 4). Gradient boosting, logistic, SVM, and random forest all substantially outperformed the baseline model. We selected gradient boosting for hyperparameter tuning based on F1 score as our metric of best performance. The decision tree model also outperformed the baseline, but overall performed worse than the other three models.

#### 4.2. Hyperparameter Tuning

After selecting the best performing regression and classification models, we then performed hyperparameter tuning using both Grid Search and Random Search on each in order to compare modalities and runtimes (Table 5). Interestingly, gradient boosting regressor was the only model which showed a meaningfully faster random search than grid search. Random search for random forest regression took almost twice as much time as grid search did for the same model, and both tuning methods for the gradient boosting classifier were quite similar.

For gradient boosting regression we saw the biggest improvement using Grid Search, which resulted in  $R^2 = 0.4322$  from the base gradient boosting  $R^2 = 0.4056$ , as

compared to Random Search which resulted in a decrease in performance, as shown by  $R^2 = 0.3949$  (Figure 7a, Table 6).

For random forest regression, both methods resulted in improved performance over the base. Grid search resulted in  $R^2 = 0.4759$  and random search resulted in  $R^2 = 0.4804$  from the base random forest  $R^2 = 0.4689$ . Within the regression models on which we performed hyperparameter tuning, random search on random forest regression yielded the highest  $R^2$ . As a result, we considered this model our highest performing regression model for further metrics (Figure 7b, Table 6).

For gradient boosting classification, both methods resulted in improved performance over the base. Grid search resulted in ROC-AUC = 0.8302 and random search resulted in ROC-AUC = 0.8358 from the base gradient boosting ROC-AUC = 0.8226 (Figure 7c, Table 7). The three gradient boosting models maintain F1 scores above the other models with default parameters, and maintain ROC curves in accordance with this assessment (Figure 12). Keeping in mind that we only performed hyperparameter tuning on one classification model, random search on gradient boosting classification yielded the highest ROC-AUC. As a result, we considered this model our highest performing classification model for further metrics.

#### 4.3. Feature Importance

Feature importance analysis helps us identify which variables contributed the most to model predictions. Figure 5 shows feature importance for the gradient boosting model after hyperparameter tuning. Housing units, percentage single parent, percentage of White individuals, and percentage of vacant housing were among the most influential predictors.

Similarly, Figure 6 shows feature importance for the random forest model after hyperparameter tuning. Retirement destination status, housing units, percentage of White individuals, and percentage of vacant housing were among the most influential predictors. The similarity in feature importance across both models suggests that there are consistent relationships between housing, demographic, and economic characteristics and migration trends.

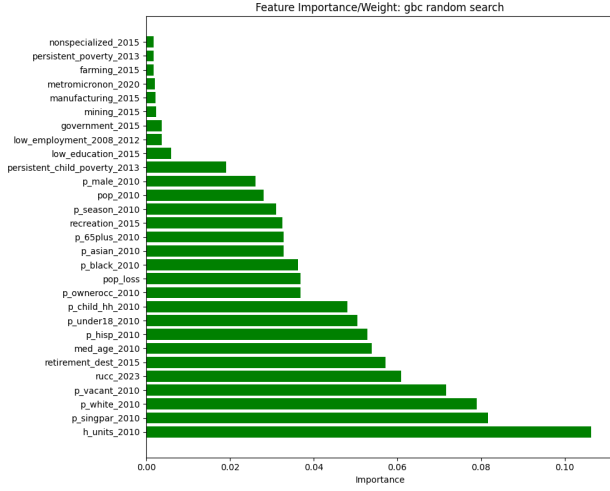


Figure 5. GBC Random

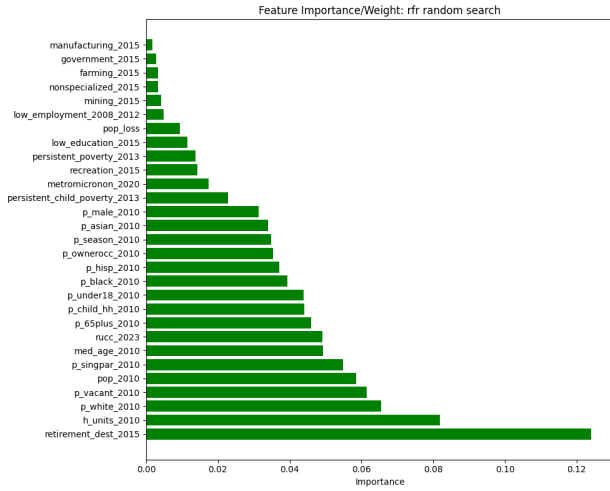


Figure 6. RFR Random

Table 5. Hyperparameter Search Runtimes (seconds)

| Model                        | Search Type | Time (s) |
|------------------------------|-------------|----------|
| Gradient Boosting Regressor  | Grid        | 10.55    |
| Gradient Boosting Regressor  | Random      | 10.03    |
| Random Forest Regressor      | Grid        | 6.85     |
| Random Forest Regressor      | Random      | 12.10    |
| Gradient Boosting Classifier | Grid        | 9.56     |
| Gradient Boosting Classifier | Random      | 9.63     |

## 5. Discussion

Our goal was to build useful regression and classification models to predict net-migration to counties based on demographic and economic features. In order to do so, we trained regression models: Linear Regression, Ridge Regression,

Table 6. Regression Model Performance (Including Hyperparameter-Tuned Models)

| Model                    | $R^2$         | MSE          |
|--------------------------|---------------|--------------|
| RFR Random Search        | <b>0.4804</b> | <b>36.24</b> |
| RFR Grid Search          | 0.4759        | 36.55        |
| Random Forest (base)     | 0.4689        | 37.04        |
| GBR Grid Search          | 0.4322        | 39.60        |
| Gradient Boosting (base) | 0.4056        | 41.46        |
| Linear Regression        | 0.3952        | 42.18        |
| GBR Random Search        | 0.3949        | 42.20        |
| Ridge Regression         | 0.3946        | 42.22        |
| SGD Regression           | 0.3830        | 43.03        |
| Elastic Net              | 0.3590        | 44.70        |
| Lasso Regression         | 0.3499        | 45.34        |
| Baseline                 | -0.0002       | 69.76        |
| Decision Tree            | -0.1112       | 77.50        |

Table 7. Classification Performance (Including Hyperparameter-Tuned Models)

| Model         | F1            | Acc.          | Prec.         | Recall        | AUC           |
|---------------|---------------|---------------|---------------|---------------|---------------|
| SVM           | 0.7201        | 0.7492        | 0.7319        | 0.7087        | <b>0.8347</b> |
| GBC Rand.     | <b>0.7435</b> | 0.7652        | 0.7396        | <b>0.7474</b> | 0.8358        |
| RF Base       | 0.7344        | 0.7620        | <b>0.7464</b> | 0.7228        | 0.8334        |
| GBC Grid      | 0.7435        | 0.7652        | 0.7396        | 0.7474        | 0.8302        |
| GBC Base      | 0.7435        | 0.7652        | 0.7396        | 0.7474        | 0.8226        |
| Logistic      | 0.7402        | <b>0.7668</b> | 0.7509        | 0.7298        | 0.8179        |
| Decision Tree | 0.6360        | 0.6709        | 0.6406        | 0.6316        | 0.6677        |
| Baseline      | 0.0000        | 0.5447        | 0.0000        | 0.0000        | 0.5000        |

Lasso Regression, SGD Regression, Random Forest, Gradient Boosting, Decision Tree, and Elastic Net. We selected Gradient Boosting and Random Forest as the highest performing models and ran hyperparameter tuning on them, comparing both grid search and random search methods. Tuned Random Forest produced the highest  $R^2$  value of 0.4804. We also trained the classification models: Decision Tree, Random Forest, Logistic Regression, SVM, and Gradient Boosting, and we selected Gradient Boosting as the highest performing model based on F1 score. With hyperparameter tuning, this model produced an F1 score of 0.7435 and an ROC-AUC of 0.8358.

Based on the highest  $R^2$  score yielded by our tested regression models (0.4804), we can determine that the economic and demographic features included in this analysis account for a sizable amount of the variance in net migration between counties. This means that understanding these demographic and economic factors can allow us to make useful predictions about what migration trends for a given county may look like. Similarly, our best performing classification model yielded an ROC-AUC score of 0.8358, strongly improving upon the baseline model, meaning that the features analyzed here can be used to make useful predictions as to

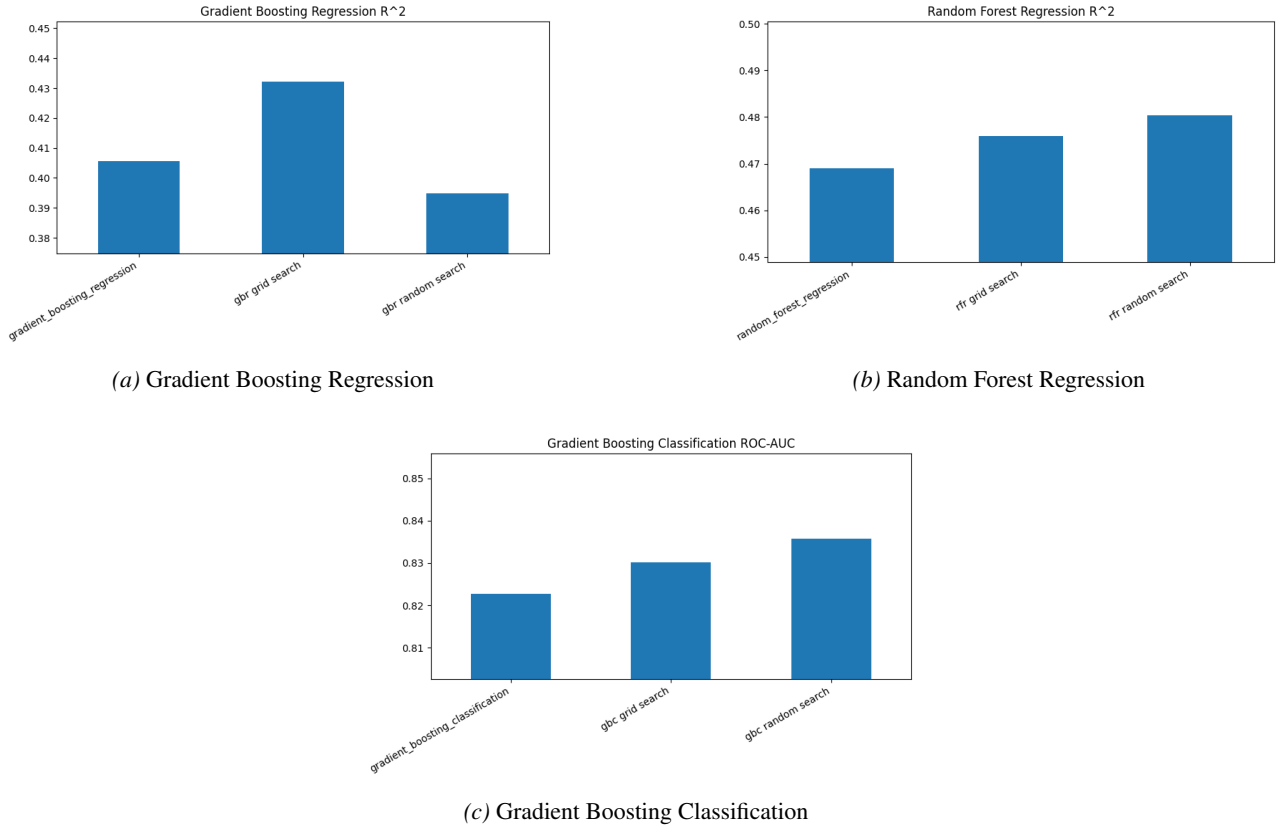


Figure 7. Hyperparameter tuning results for selected models.

whether a given county will see an influx or outward flow of migrants.

Given the complexities of modeling a human behavior like migration, we believe that the features analyzed here provide a relatively strong basis on which to predict. Additional factors, such as policy, climate, and geography, likely also play an influential role in motivating migration to and from given counties. In future iterations, expanding to these features could result in higher performing models.

The performance of each type of model helps to elucidate the kinds of relationships present in the data. The strongest performing regression models, Random Forest and Gradient Boosting, were ensemble methods, more capable of capturing complex and non-linear relationships in the features. This is likely because people migrate due to a combination of factors. Some of these factors may act more like thresholds, while other relationships may be more linear, and therefore ensemble methods are better able to capture these relationships. Unregularized Linear Regression, Ridge Regression, and SGD Regression yielded lower  $R^2$  scores, further indicating that relationships between the analyzed features and the target variable cannot always be assumed to be linear, though these models still have predictive value. Lasso Regression and Elastic Net underperformed the Un-

regularized Linear Regression, signaling that reducing lower weighted features to zero reduces the performance of the model. This suggests that these features are still necessary to the model.

Both the ensemble classification models (SVM, Random Forest, and Gradient Boosting) and the linear classification model (Logistic Regression) performed similarly strongly. This suggests that while capturing the specific magnitude of net migration requires modeling complex, non-linear feature interactions, the binary task of determining whether a county will experience a net inflow or outflow is more straightforward and can be modeled faithfully with linear decision boundaries.

Among both regression and classification tasks, the Decision Tree models were the weakest. This indicates that Decision Tree's rectangular decision boundaries are insufficient for capturing the patterns that drive migration. Unlike linear models, which can account for gradual shifts, the discrete partitioning of a decision tree appears to oversimplify the underlying migration dynamics.

Hyperparameter tuning yielded improved performance above the original model for all three models we selected. Grid search proved to be the most effective hyperparameter

tuning method for gradient boosting regression, yielding  $R^2 = 0.4322$  over the original  $R^2 = 0.4056$ . Notably, the random search tuned model performed worse than the original model which could point to either a failure to find good hyperparameters within the maximum iterations or finding repeatedly bad parameters by chance. For the random forest regression model and gradient boosting classification, both grid search and random search outperformed the default parameter model, with random search yielding the highest  $R^2$ . This could be due to random search being a less confined approach than grid search, and therefore finding better hyperparameters.

Among the tested regression models, the residuals are greater for more extreme predictions (Figure 9, 10). This is likely because net migration numbers around zero are most common within the training data. For the rarer higher magnitude migration numbers, all models show higher residuals. An additional effect that can be seen is that the predicted net migration values tend to be closer to zero than the actual values. This is most pronounced on the worse performing models (Lasso and Elastic Net), and less pronounced on the best models (Random Forest and Gradient Boosting).

Among the tested classification models, we found that precision and recall are closely balanced across all models, meaning that errors are fairly evenly distributed between false negatives and false positives. By comparing the mean feature values of correctly and incorrectly classified counties by our top classification model (tuned Gradient Boost) we can see that more populous counties were more likely to be misclassified (Table 8). Additionally, counties that are labeled as retirement destinations were far more likely to be correctly classified than others, suggesting that while the model has learned that retirement destinations are likely to see an influx of migrants, counties that do not have this designation may be harder for the model to classify.

For our highest performing regression model (tuned Random Forest) whether or not a county is a retirement destination was given the highest weight, this was followed by number of housing units, percent White population, percent of homes that are vacant, and total population. Our highest performing classification model (tuned Gradient Boost) had a similar but distinct breakdown, with number of housing units being assigned the highest weight, followed by percent single parent households, percent White, percent of homes vacant, and rural-urban continuum classification. Overall, these features are weighed heavily in all of the models we ran. This suggests that both the classification models and the regression models are using similar features to make predictions. This also gives us insight into what key considerations are for the movement of people. Based on the importance of housing unit totals and vacancies, it seems apparent that housing concerns are a driver of migration.

Additionally, whether a county is a retirement destination ranks as the most important feature in many of our models, meaning that retirement is likely to be another key driver of migration. Surprisingly, some economic factors, such as whether a county is mining or manufacturing dependent, are not given high importance by most models, indicating that these features are less of a consideration in migration than one might expect.

## 5.1. Conclusion

Overall, we were able to produce models that are useful in predicting net migration for each U.S. county. Gradient Boosting (regression and classification) and Random Forest (regression) models produced the strongest performance according to our chosen metrics. Our experiments yield a set of feature weights that can be analyzed to provide insight into which economic and demographic factors affect migration patterns in the United States.

While our model demonstrates substantial predictive power, including a more diverse set of features could improve model performance. Further work could incorporate additional climate, geographical, and political features, as well as relationships between counties to further improve the predictive performance and better capture migration patterns in the United States. We also selected which models to perform hyperparameter tuning on according to the metrics that seemed most relevant according to our evaluation. However, further study could include a more comprehensive analysis of which metric provides the true best performance, and all models could be tuned and compared. Further implementation could also include k fold cross validation in order to reduce reliance on our specific train and test splits.

## References

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## 6.2. Model Diagnostics

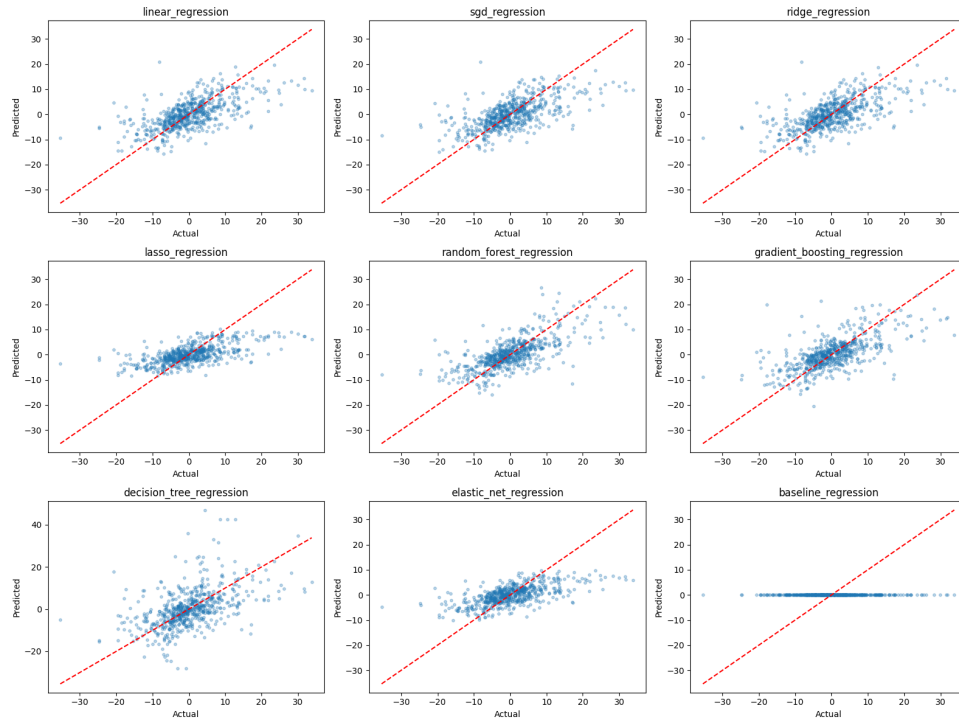


Figure 9. Actual vs Predicted Scatter Plots for Regression

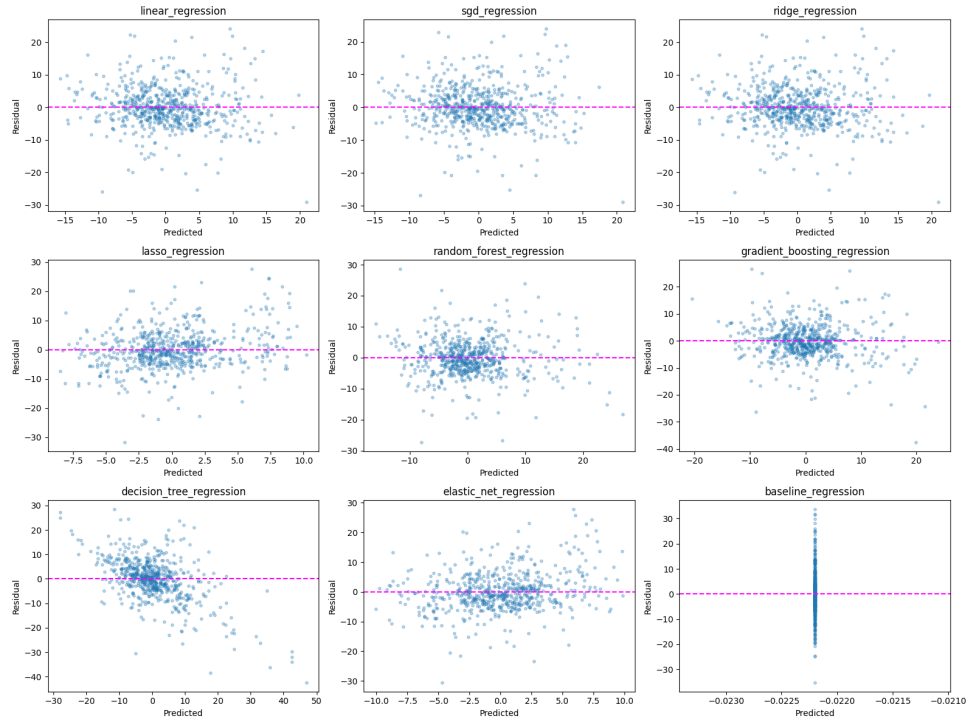


Figure 10. Predicted vs Residual for Regression

### 6.3. Feature Importance (After Hyperparameter Tuning)

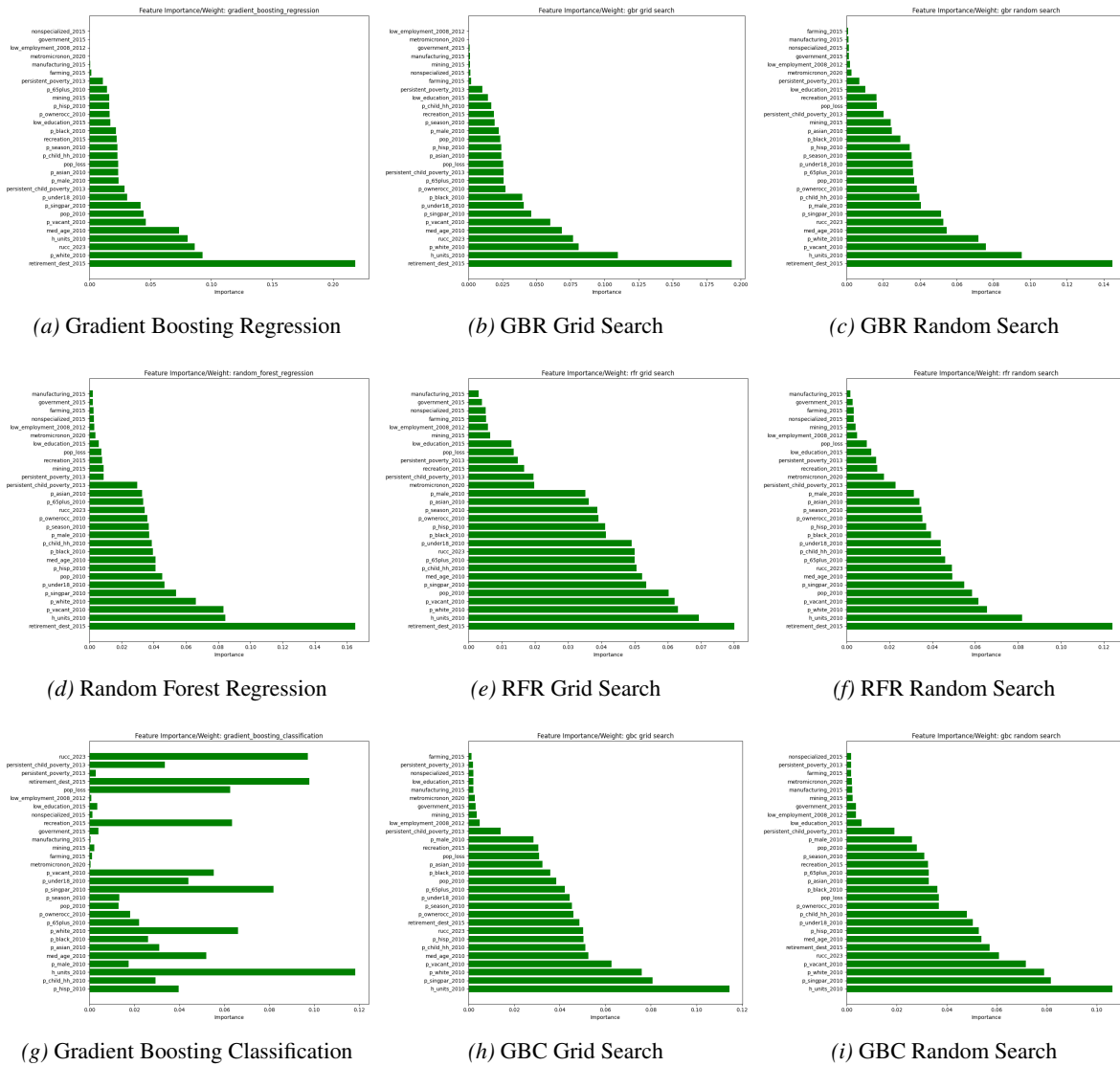


Figure 11. Feature importance and model weights across hyperparameter tuned models

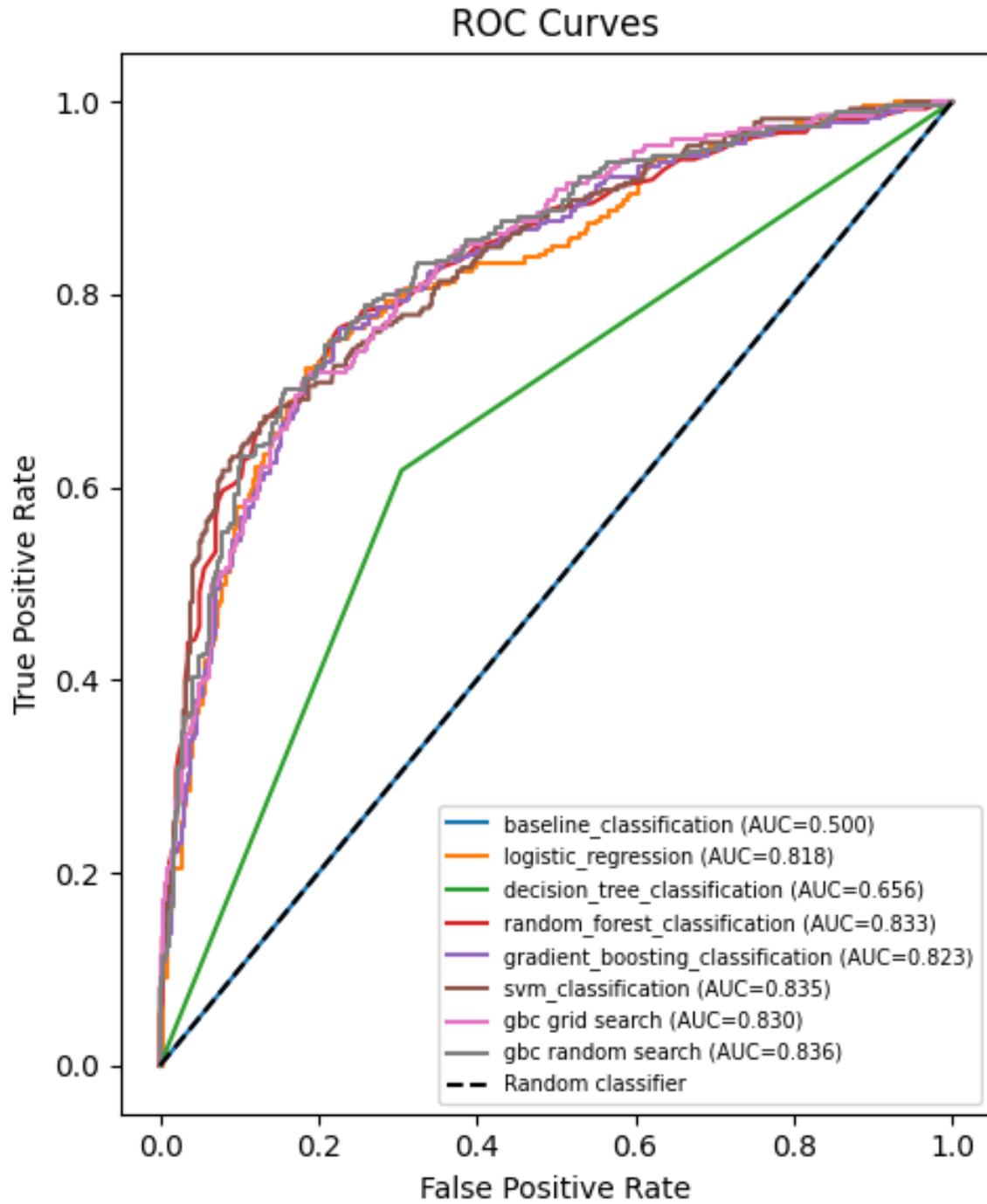


Figure 12. ROC curves for classification models after hyperparameter tuning.

## 6.4. Feature Means for Correctly and Incorrectly Classified Counties

Table 8. Comparison of Feature Means: Correct vs. Failed Classification Predictions

| Feature Name               | Correctly Classified | Failed Predictions | Overall Mean |
|----------------------------|----------------------|--------------------|--------------|
| % Hispanic (2010)          | 7.861                | 6.807              | 8.297        |
| % Households w/ Children   | 31.571               | 30.586             | 31.522       |
| Housing Units (2010)       | 40,132.85            | 46,676.07          | 41,648.50    |
| % Male                     | 50.026               | 49.986             | 49.971       |
| Median Age (2010)          | 40.382               | 40.743             | 40.342       |
| % Asian                    | 0.991                | 1.110              | 1.132        |
| % Black                    | 9.463                | 8.238              | 8.773        |
| % White                    | 78.365               | 80.852             | 78.366       |
| % 65 Plus                  | 15.815               | 16.308             | 15.901       |
| % Owner Occupied           | 72.631               | 72.346             | 72.267       |
| Total Population (2010)    | 90,492.41            | 114,693.73         | 97,607.06    |
| % Seasonal Housing         | 7.126                | 8.493              | 6.742        |
| % Single Parent            | 8.897                | 8.309              | 8.791        |
| % Under 18                 | 23.365               | 22.937             | 23.416       |
| % Vacant Housing           | 9.580                | 9.525              | 9.540        |
| Metro/Micro/Non-core       | 2.071                | 2.095              | 2.038        |
| Farming Dependent (2015)   | 0.140                | 0.170              | 0.162        |
| Mining Dependent (2015)    | 0.084                | 0.054              | 0.082        |
| Manufacturing Dependent    | 0.171                | 0.184              | 0.165        |
| Government Dependent       | 0.159                | 0.184              | 0.147        |
| Recreation Dependent       | 0.140                | 0.163              | 0.135        |
| Nonspecialized Economy     | 0.380                | 0.354              | 0.393        |
| Low Education (2015)       | 0.173                | 0.136              | 0.149        |
| Low Employment (08-12)     | 0.311                | 0.245              | 0.289        |
| Population Loss            | 0.186                | 0.150              | 0.169        |
| Retirement Dest. (2015)    | 0.186                | 0.068              | 0.141        |
| Persistent Poverty (2013)  | 0.134                | 0.061              | 0.112        |
| Pers. Child Poverty (2013) | 0.251                | 0.204              | 0.226        |
| RUCC (2023)                | 5.299                | 5.571              | 5.247        |